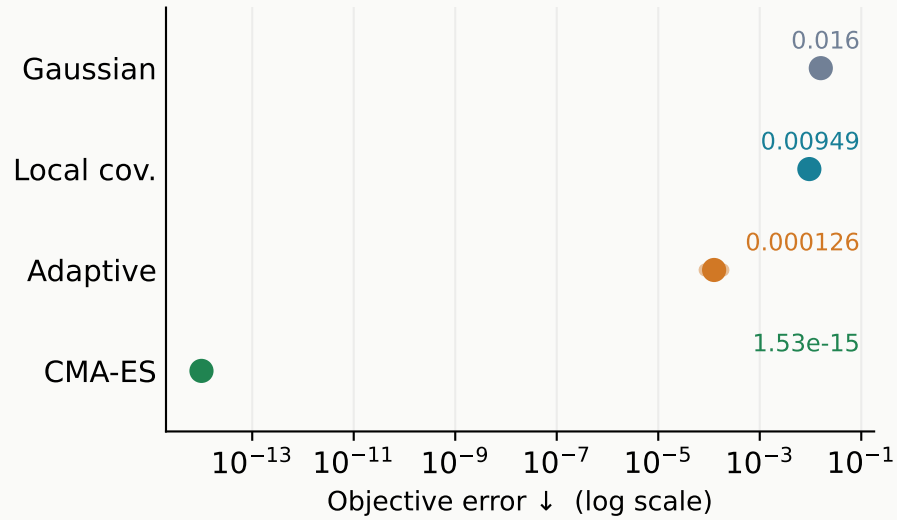


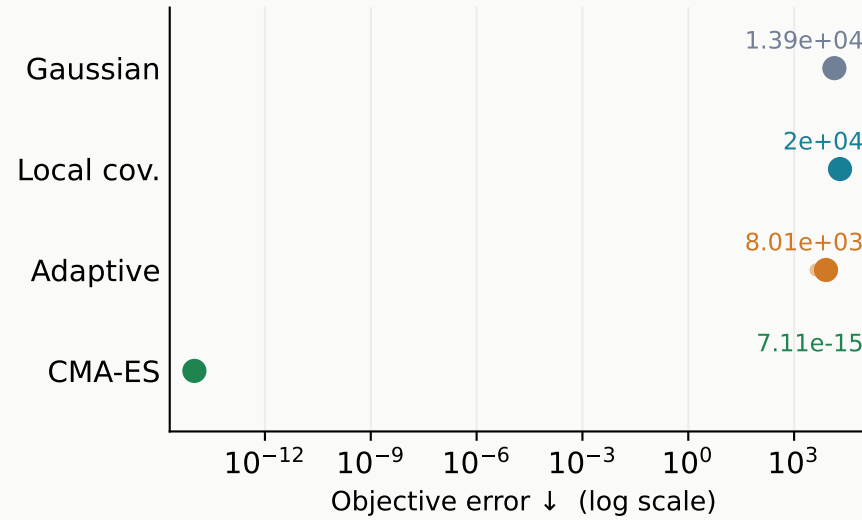
Fractal movement strategies vs CMA-ES

20 dimensions · 100,000-evaluation cap · 10 seeds · Wave: 128 walkers, 5 elites

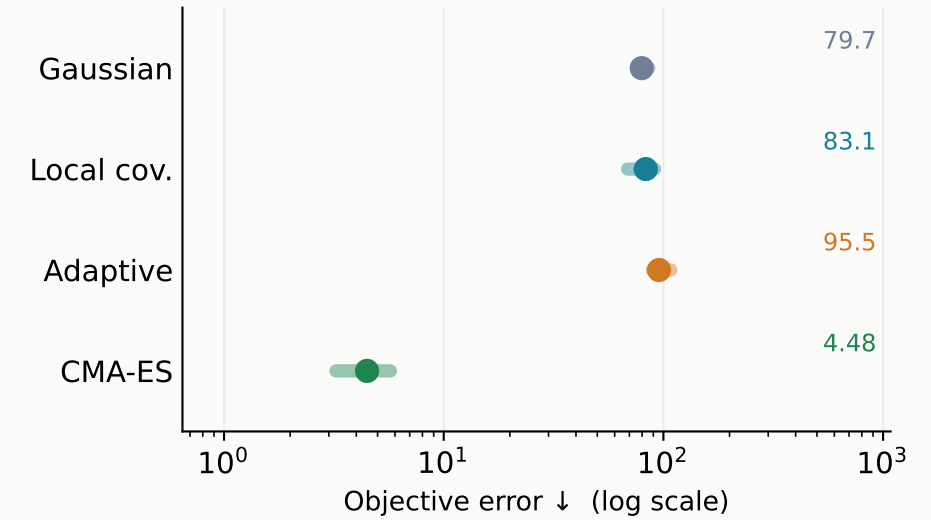
Quadratic bowl



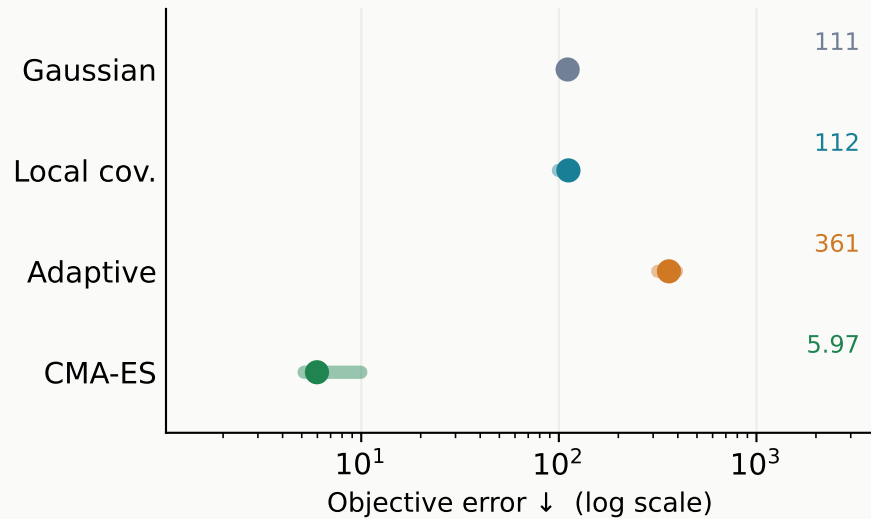
Rotated ellipsoid



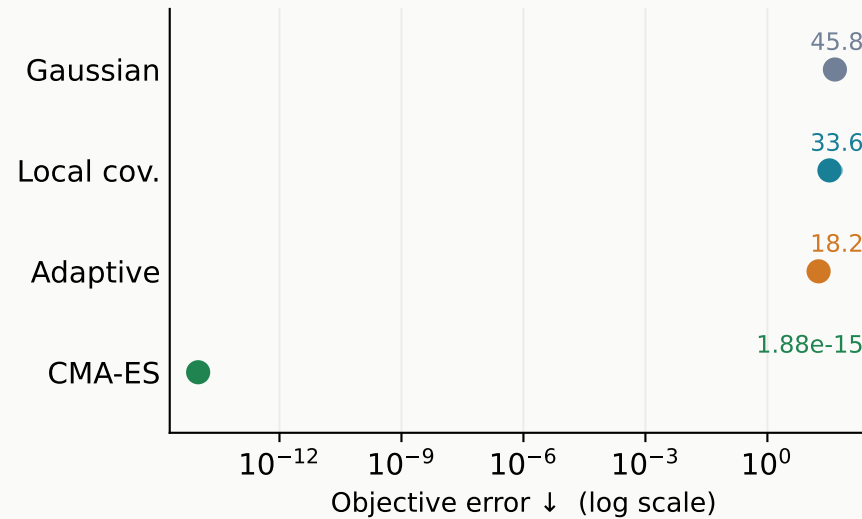
Rastrigin



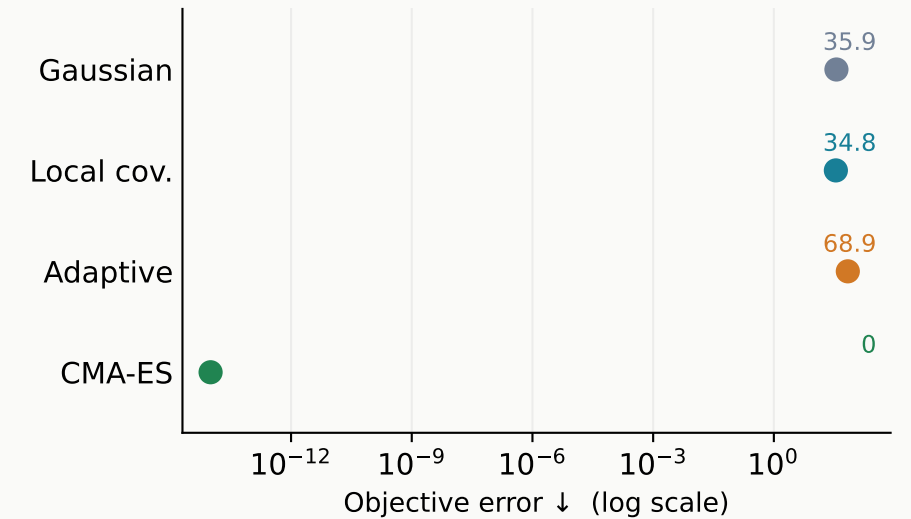
Rotated Rastrigin



Rosenbrock



Boundary optimum



Dots: median · thick intervals: middle 50% of seeds · printed values: exact medians (rounded)

Errors below 10^{-14} are drawn at 10^{-14} . CMA-ES uses its own population. Controller-enabled adaptive results overlap exactly: no restarts.